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COMPUTE-ON-FAN-TRAY-ASSEMBLY (COFTA) SYSTEM

Abstract

A Compute-On-Fan-Tray-Assembly (COFTA) system includes a fan chassis that is configured to be positioned in a networking device, and that has a fan module coupling subsystem that is configured to couple at least one fan module to the fan chassis. A compute subsystem is mounted to the fan chassis and is configured to perform compute functionality for the networking device. A networking device connector is included on the fan chassis, is coupled to the compute subsystem, and is configured to engage a compute subsystem connector in the networking device to couple the compute subsystem to a network processing system in the networking device when the fan chassis is positioned in the networking device.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates generally to information handling systems, and more particularly to a Compute-On-Fan-Tray-Assembly (COFTA) system for use with information handling systems.

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

[0003] Information handling systems such as, for example, networking devices such as switches, utilize a Central Processing Unit (CPU) or other compute device in addition to their Network Processing Unit (NPU) in order to provide an operating system (e.g., a Network Operating System (NOS)) for the networking device, perform management operations for the networking device, and/or provide other functionality (e.g., non-data plane functionality) that is not performed by the NPU. For example, conventional networking devices are often provided with a Compute-On-Module (COM) Express device that includes a CPU and that is mounted to a motherboard in the networking device (i.e., using “vertical”/“Z-axis” connectors that connect to corresponding “vertical”/“Z-axis” connectors on the motherboard to position the COM Express device above and parallel to the motherboard) in order to allow that CPU to perform the functionality for the networking device discussed above. However, the use of COM Express devices to provide the CPU in networking devices raises some issues.

[0004] For example, COM Express devices may require servicing or replacement for a variety of reasons that include CPU errors, storage device degradation that can prevent the booting of the OS, memory device issues, and/or other COM Express device issues that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, in many situations, the upgrade (or downgrade) of the COM Express device may be desirable, such as when the NOS requirements for the networking device have changed, storage requirements for the networking device have changed, processing requirements for the networking device have changed, security requirements for the networking device have changed, etc.

[0005] However, the COM Express devices discussed above are not designed to be field replaceable. Rather, in order to remove a COM Express device from its networking device, that networking device must be removed from its rack, have its top cover removed (an action that involves the removal of dozens of screws and operates to void the manufacturer warranty in most situations if not performed by an authorized entity), and then the COM Express device must be unscrewed from the motherboard in that networking device. As such, when a COM Express device requires the servicing or replacement described above, its networking device is removed from its rack and that networking device is shipped to an authorized COM Express device servicing/replacement entity, which results in networking device downtime and can be particularly costly as shipping carriers typically use volumetric weighting. Furthermore, the current solution to “upgrading” networking device COM Express devices is to buy a newer generation networking

device.
[0006] Accordingly, it would be desirable to provide a compute subsystem in a networking device in a manner that addresses the issues discussed above.
SUMMARY
[0007] According to one embodiment, an Information Handling System (IHS) includes a networking device chassis that defines at least one fan slot; a network processing system that is housed in the networking device chassis and that is configured to perform networking functionality; a fan chassis that includes at least one fan module and that is positioned in the at least one fan slot defined by the networking device chassis; a compute subsystem that is mounted to the fan chassis and that is configured to perform compute functionality; and a networking device connector that is included on the fan chassis, that is coupled to the compute subsystem, that engages a compute subsystem connector in the networking device chassis to couple the compute subsystem to the network processing system, and that is configured to disengage the compute subsystem connector when the fan chassis is removed from the networking device chassis to decouple the compute subsystem from the network processing system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a schematic view illustrating an embodiment of an Information Handling System (IHS).
[0009] FIG. **2**A is a schematic view illustrating an embodiment of a COFTA system that may be provided according to the teachings of the present disclosure.
[0010] FIG. **2**B is a perspective view illustrating an embodiment of the COFTA system of FIG. **2**A.
[0011] FIG. **3**A is a schematic view illustrating an embodiment of a networking device that may be used with the COFTA system of FIGS. **2**A and **2**B.
[0012] FIG. **3**B is a perspective view illustrating an embodiment of a motherboard in the networking device of FIG. **3**A.
[0013] FIG. **4** is a flow chart illustrating an embodiment of a method for providing a compute system in a networking device.
[0014] FIG. **5**A is a perspective view illustrating an embodiment of a plurality fan modules being coupled to the COFTA system of FIGS. **2**A and **2**B.
[0015] FIG. **5**B is a perspective view illustrating an embodiment of a plurality fan modules coupled to the COFTA system of FIGS. **2**A and **2**B.
[0016] FIG. **5**C is a schematic view illustrating an embodiment of a plurality fan modules coupled to the COFTA system of FIGS. **2**A and **2**B.
[0017] FIG. **6** is a perspective view illustrating an embodiment of a plurality fan modules and the COFTA system of FIGS. **5**B and **5**C being coupled to the networking device of FIGS. **3**A and **3**B.
[0018] FIG. **7**A is a schematic view illustrating an embodiment of a plurality fan modules coupled to the networking device of FIGS. **3**A and **3**B.
[0019] FIG. **7**B is a rear view illustrating an embodiment of a plurality fan modules coupled to the networking device of FIGS. **3**A and **3**B.
[0020] FIG. **8** is a schematic view illustrating an embodiment of the COFTA system of FIGS. **5**B and **5**C coupled to the networking device of FIGS. **7**A and **7**B.
[0021] FIG. **9** is a schematic view illustrating an embodiment of the COFTA system of FIGS. **2**A and **2**B and the networking device of FIGS. **3**A and **3**B operating during the method of FIG. **4**.
[0022] FIG. **10** is a schematic view illustrating an embodiment of the COFTA system of FIGS. **2**A and **2**B and the networking device of FIGS. **3**A and **3**B operating during the method of FIG. **4**.
DETAILED DESCRIPTION

[0023] For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0024] In one embodiment, IHS **100**, FIG. **1**, includes a processor **102**, which is connected to a bus **104**. Bus **104** serves as a connection between processor **102** and other components of IHS **100**. An input device **106** is coupled to processor **102** to provide input to processor **102**. Examples of input devices may include keyboards, touchscreens, pointing devices such as mouses, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device **108**, which is coupled to processor **102**. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS **100** further includes a display **110**, which is coupled to processor **102** by a video controller **112**. A system memory **114** is coupled to processor **102** to provide the processor with fast storage to facilitate execution of computer programs by processor **102**. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis **116** houses some or all of the components of IHS **100**. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor **102** to facilitate interconnection between the components and the processor **102**.

[0025] Referring now to FIGS. **2**A and **2**B, an embodiment of a Compute-On-Fan-Tray-Assembly (COFTA) system **200** is illustrated that may be provided according to the teachings of the present disclosure. In the illustrated embodiment, the COFTA system **200** includes a fan chassis **202** that supports the components of the COFTA system **200**. As will be appreciated by one of skill in the art in possession of the present disclosure, the fan chassis **202** of the COFTA system **200** may be considered a “fan tray assembly” that is configured support one or more fan modules, as well as configured to be positioned in a networking device (or other computing device) to couple those fan module(s) to the networking device (or other computing device) in order to allow those fan modules to be used to provide cooling for the networking device (or other computing device). As such, the fan chassis **202** may include any of a variety of device coupling features that one of skill in the art in possession of the present disclosure would appreciate allow the coupling of the fan chassis **202** to the networking device as described below.

[0026] In the illustrated embodiment, the fan chassis **202** is configured to support three fan modules, and includes a fan coupling subsystem that includes fan slots **204***a*, **204***b*, and **204***c* located adjacent a rear edge of the fan chassis **202**. As will be appreciated by one of skill in the art in possession of the present disclosure, each of the fan slots **204***a*-**204***c* may include any of a variety of fan module coupling features that are configured to couple the fan modules to the fan chassis **202** as described below. However, while the specific examples provided herein describe a fan chassis **202** with a fan coupling subsystem configured to support three fan modules, one of skill in the art in possession of the present disclosure will appreciate how the fan chassis **202** may be configured to support fewer fan modules (including a single fan module) or more fan modules while remaining within the scope of the present disclosure as well. For example, the number of fan modules supported by the fan chassis **202** may be dictated by the width of the circuit board required to support the components of the compute subsystem **210** (discussed in further detail below).

[0027] The fan chassis **202** may also be provided with a respective fan module connector for each fan module it is configured to support, and thus includes a fan module connector **206***a*, **206***b*, and **206***c* located adjacent each of the fan slots **204***a*, **204***b*, and **204***c*, respectively, in the embodiments illustrated and described below. The fan chassis **202** is also provided with a fan controller connector **208** that is coupled to each of the fan module connectors **206***a*-**206***c*, and as described below the fan controller connector **208** may be configured to aggregated signals received from fan modules via the fan module connectors **206***a*-**206***c*, as well as split signals received for fan modules connected to the fan module connectors **206***a*-**206***c*.

[0028] As illustrated, a compute subsystem **210** may be mounted to the fan chassis **202**. For example, the compute subsystem **210** may include a circuit board **211** that, as can be seen in FIG. **2**B, may be mounted to the fan chassis **202** via a plurality of stand-offs **211***a* (i.e., the two stand-offs **211***a* visible in FIG. **2**B engaging the circuit board **211** adjacent its corners along a common edge, as well as two stand-offs that are not visible in FIG. **2**B but that may engage the circuit board **211** opposite the circuit board **211** from the visible stand-offs **211***a* adjacent its corners along a common opposing edge) in order to provide the circuit board **211** in a spaced-apart orientation along a Z-axis from the adjacent surface of the fan chassis **202** in order to, for example, provide additional thermal cooling for the compute subsystem **210** by allowing airflow from fan modules (described in further detail below) to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **2**B) of the circuit board **211**.

[0029] As will be appreciated by one of skill in the art in possession of the present disclosure, the compute subsystem **210** may include any of a variety of compute components. For example, in the embodiments illustrated and discussed below, the compute subsystem **210** includes a primary processing system provided by a Central Processing Unit (CPU) **212***a* that is coupled to storage devices provided by a Serial Peripheral Interface (SPI) flash storage device **212***c* and an Solid State Drive (SSD) storage device **212***c*, memory devices provided by Dynamic Random Access Memory (DRAM) device(s) **212***d* and an Electronically Erasable Programmable Read-Only Memory (EEPROM) device **212***e*, a security device provided by a Trusted Platform Module (TPM) device, and a secondary processing device provided by a Complex Programmable Logic Device (CPLD) **212***e*. However, while specific processing devices, storage devices, memory devices, and security devices are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem **210** may include any of a variety of processing devices, storage devices, memory devices, security devices, and/or other devices based on the compute functionality required from the compute subsystem **210** for any particular COFTA application.

[0030] As can be seen in FIGS. **2**A and **2**B, the compute subsystem **210** includes a networking device connector **214** that, in the illustrated example, is coupled to the CPU **212***a*. In a specific embodiment, the compute subsystem **210** may be an adaptation of a Compute-On-Module (COM) Express device that includes a modified connector, and thus may be provided with a COM Express form-factor with the exception of that modified connector. For example, the networking device connector **214** may be configured similarly to a COM Express device connector, with the exception that the networking device connector **214** is oriented, and configured to connect along, an X-axis (i.e., the axis parallel to the circuit board **210** and the fan chassis **202**) rather than an Z-axis (i.e., the axis perpendicular to the circuit board **210** and the fan chassis **202**) that conventional COM Express device connectors are oriented and configured to connect along. However, while a specific compute subsystem and networking device connector are illustrated and described, one of skill in the art in possession of the present disclosure will appreciate how other compute subsystems and networking device connectors will fall within the scope of the present disclosure as well. Furthermore, while a specific COFTA system **200** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that the COFTA system of the present disclosure may include a variety of components and component configurations while remaining within the scope of the present disclosure as well.

[0031] Referring now to FIGS. **3**A and **3**B, an embodiment of a networking device **300** is illustrated that may be used with the COFTA system of FIGS. **2**A and **2**B. In an embodiment, the networking device **300** may be provided by the IHS **100** discussed above with reference to FIG. **1** and/or may include some or all of the components of the IHS **100**, and in specific examples may be provided by a switch device. However, while illustrated and discussed as being provided by a switch device, one of skill in the art in possession of the present disclosure will recognize that the functionality of the networking device **300** discussed below may be provided by other devices that are configured to operate similarly as discussed below. For example, one of skill in the art in possession of the present disclosure will appreciate how the COFTA system of the present disclosure may be utilized in a server device similarly as COM Express devices are utilized with relatively “low-end” (or otherwise relatively inexpensive) server devices to provide the CPUs in those server devices. As such, in some embodiments of the present disclosure the networking device **300** may be replaced by a server device, with components in that server device replacing the network processing system **318** and being coupled to the compute subsystem connector **314** discussed below.

[0032] In the illustrated embodiment, the networking device **300** includes a device chassis **302** that houses the components of the networking device **300**, only some of which are illustrated and described below. For example, the device chassis **302** may include a fan coupling subsystem that includes fan slots **304***a*, **304***b*, **304***c*, and **304** that are located adjacent a rear wall of the device chassis **302**. As will be appreciated by one of skill in the art in possession of the present disclosure, each of the fan slots **304***a*-**204***c* may include any of a variety of fan module coupling features that are configured to couple the fan modules to the device chassis **302** as described below.

[0033] Furthermore, the device chassis **302** may also include a COFTA coupling subsystem that includes a COFTA slot **306** that one of skill in the art in possession of the present disclosure will appreciate operates to provide one or more additional fan slots (i.e., **3** additional fan slots in the specific examples provided below) on the device chassis **302** that are located adjacent the rear wall of the device chassis **302** (i.e., the COFTA system **200** is configured to be positioned in fan slot(s) on the device chassis **302** that provide the COFTA slot **306**). As such, one of skill in the art in possession of the present disclosure will appreciate how the COFTA slot in networking devices described below may be provided in place of some number of fan slots that would otherwise be provided in a conventional version of those networking devices. As will be appreciated by one of skill in the art in possession of the present disclosure, the COFTA slot **306** may include any of a variety of COFTA system coupling features that are configured to couple the fan chassis **202** in the COFTA system **200** to the device chassis **302** as described below.

[0034] However, while the specific examples provided herein describe a device chassis **302** with a fan coupling subsystem and COFTA coupling subsystem that is configured to support seven fan modules, one of skill in the art in possession of the present disclosure will appreciate how the device chassis **302** may be configured to support fewer fan modules (including a single fan module) or more fan modules while remaining within the scope of the present disclosure as well.

[0035] As illustrated, the device chassis **302** may house a motherboard **308** that is configured to support the components of the networking device **300**. In the specific examples provided herein, the motherboard **308** defines a COFTA channel **308***a* that provides the motherboard **308** with a “U” shape that one of skill in the art in possession of the present disclosure will appreciate is configured to allow the COFTA system **200** of FIGS. **2**A and **2**B to be moved into and positioned in the device chassis **302**. The motherboard **308** may also be provided with fan module connectors for each fan module that the networking device **300** is configured to support, and thus includes a fan module connector **310***a*, **310***b*, **310***c*, and **310***d* located adjacent each of the fan slots **304***a*, **304***b*, **304***c*, and **304***d*, respectively, as well as a fan subsystem connector **312** located adjacent the COFTA channel **308***a* in the embodiments illustrated and described below. The motherboard **308** also includes a compute subsystem connector **314** that is located adjacent the COFTA channel **308**, and one of skill in the art in possession of the present disclosure will appreciate how the compute subsystem connector **314** may be elevated from the adjacent surface of the motherboard **308** (e.g., in the Z-axis) relative to the fan subsystem connector **312** (as can be seen in FIG. **3**B) in embodiments in which the circuit board **211** and networking device connector **214** on the compute subsystem **210** are elevated as illustrated in FIG. **2**B.

[0036] A fan controller subsystem **316** is mounted to the motherboard **308** and is coupled (e.g., via traces in the motherboard **308**) to each of the fan module connectors **310***a*-**310***d* and the fan subsystem connector **312**. A network processing system **318** is also mounted to the motherboard **308** and connected (e.g., via a network processing system connector **318***a*, visible in FIG. **3**B, and traces in the motherboard **308**) to each of the fan controller subsystem **316** and the compute system connector **314**. As discussed above, in some embodiments the network processing system **318** may be provided by a Network Processing Unit (NPU). However, as illustrated in FIG. **3**B, the network processing system connector **318***a* may additionally) connect other processing systems (e.g., a Field Programmable Gate Array (FPGA) device, a CPLD, etc.), a Baseboard Management Controller (BMC) device, a front panel subsystem including a console port, an out-of-band management port, Light Emitting Devices (LEDs), and/or other systems to the compute subsystem connector **314** while remaining within the scope of the present disclosure as well. As such, while a specific networking device **300** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that networking devices (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the networking device **300**) may include a variety of components and/or component configurations for providing conventional networking functionality, as well as the COFTA functionality discussed below, while remaining within the scope of the present disclosure as well.

[0037] Referring now to FIG. **4**, an embodiment of a method **400** for providing a compute system in a networking device is illustrated. As discussed below, the systems and methods of the present disclosure provide a Compute-On-Fan-Tray-Assembly (COFTA) system that includes a compute subsystem mounted to a fan chassis that may be easily coupled to, and decoupled from, a networking device. For example, the COFTA system of the present disclosure may include a fan chassis that is configured to be positioned in a networking device, and that has a fan module coupling subsystem that is configured to couple at least one fan module to the fan chassis. A compute subsystem is mounted to the fan chassis and is configured to perform compute functionality for the networking device. A networking device connector is included on the fan chassis, is coupled to the compute subsystem, and is configured to engage a compute subsystem connector in the networking device to couple the compute subsystem to a network processing system in the networking device when the fan chassis is positioned in the networking device. As described below, embodiments of the COFTA system of the present disclosure allow the compute subsystem for the networking device to be connected to, and disconnected from, the networking device without the need to remove the networking device from its rack or suspend networking functionality performed by the network processing system in the networking device, eliminating issues with servicing and/or replacement of such compute subsystems, and enabling the upgrading (or downgrading) of such compute subsystems.

[0038] The method **400** begins at block **402** where one or more fan modules are coupled to a

COFTA system. With reference to the specific example provided in FIG. **5**A, in an embodiment of block **402**, three fan modules may be coupled to the fan chassis **202** in the COFTA system **200** of FIGS. **2**A and **2**B. In an embodiment, the fan modules **500** may be provided using configurable airflow direction fan modules like those described by the inventors of the present disclosure in U.S. patent application Ser. No. 18/479,235, attorney docket no. 133643.01, filed on Oct. 2, 2023, the disclosure of which is incorporated by reference herein in its entirety. However, one of skill in the art in possession of the present disclosure will appreciate how the use of a variety of other types of fan modules with the COFTA system **200** will fall within the scope of the present disclosure as well.

[0039] As illustrated, each of the fan modules **500** may include features for coupling that fan module to the fan chassis **202**, a corresponding fan chassis connector **500***a* that one of skill in the art in possession of the present disclosure will appreciate is coupled to a fan device in the fan module, and/or any other fan module features known in the art. At block **402**, each of the three fan modules **500** may be positioned adjacent the rear edge of the fan chassis **202** such that that fan module **500** is aligned with one of the fan slots **204***a*-**204***c*, with its fan chassis connector **500***a* located adjacent that fan slot as illustrated in FIG. **5**A. With reference to FIGS. **5**B and **5**B, each of the three fan modules **500** may then be moved towards the fan chassis **202** such that each fan module **500** engages its corresponding fan slot **204***a*-**204***c*, respectively, with continued movement of the fan modules **500** through the fan slots **204***a*-**204***c* causing their corresponding fan chassis connectors **500***a* to engage the fan module connectors **206***a*-**206***c* to couple each of those fan modules **500** to the fan controller connector **208**.

[0040] In the specific example provided herein, four of the fan modules **500** may also be coupled directly to the networking device **300** as well. For example, with reference to FIG. **6**, each of the four fan modules **500** may be positioned adjacent the rear wall of the device chassis **302** (not visible in FIG. **6**) such that that fan module **500** is aligned with one of the fan slots **304***a*-**304***d* (not visible in FIG. **6**), with its fan chassis connector **500***a* is located adjacent that fan slot. Each of the four fan modules **500** may then be moved towards the device chassis **302** such that each fan module **500** engages its corresponding fan slot **304***a*-**304***d*, respectively, with continued movement of the fan modules **500** through the fan slots **304***a*-**304***d* causing their corresponding fan chassis connectors **500***a* to engage the fan module connectors **310***a*-**310***d*, as illustrated in FIG. **7**A, to couple each of those fan modules **500** to the fan controller subsystem **316**. With reference to FIG. **7**B, the rear of the networking device **300** is illustrated with the four fan modules **500** coupled to networking device **300** as described above, and with the fan subsystem connector **312** and compute subsystem connector **314** visible via the COFTA slot **306**.

[0041] The method **400** then proceeds to block **404** where the COFTA system is coupled to a networking device. With reference back to FIG. **6**, in an embodiment of block **404**, the COFTA system **200** including the three fan modules coupled thereto as described above with reference to FIGS. **5**B and **5**C may be positioned adjacent the rear wall of the device chassis **302** (not visible in FIG. **6**) such that that COFTA system is aligned with the COFTA slot **306** (not visible in FIG. **6**), with its fan controller connector **208** and networking device connector **214** located adjacent the COFTA slot **306**. The COFTA system **200** may then be moved towards the device chassis **302** such that it engages the COFTA slot **306**, with continued movement of the COFTA system **200** through the COFTA slot **306** causing the COFTA system **200** to move through the COFTA channel **308***a* defined by the motherboard **308** until its fan controller connector **208** and networking device connector **214** engage the fan subsystem connector **312** and the compute subsystem connector **314**, respectively, on the motherboard **308** in the networking device **300**, as illustrated in FIG. **8**, to couple each of the fan modules **500** on the COFTA system **200** to the fan controller subsystem **316**, and couple the compute subsystem **210** to the network processing system **318**.

[0042] The method **400** then proceeds to block **406** where the fan modules are operated to cool the networking device. With reference to FIG. **9**, in an embodiment of block **406**, the fan controller subsystem **316** may perform fan control operations **900** that include using any of a variety of fan control techniques known in the art to control the fan modules **500** coupled to the fan module connectors **310***a*-**310***d* on the motherboard **308**, as well as to control the fan modules **500** coupled to fan subsystem connector **312** on the motherboard **308** (e.g., via the connection of the fan controller connector **208** on the COFTA system **200** to the fan subsystem connector **312**, and the coupling of the fan module connectors **206***a*-**206***c* on the COFTA system **200** to the fan controller connector **208**), in order to operate any of the fan modules **500** to produce an airflow through the device chassis **302** to cool any components housed in the device chassis **302** (e.g., the network processing system **318**, the compute subsystem **210**, as well as any of a variety of unillustrated components that would be apparent to one of skill in the art in possession of the present disclosure). As will be appreciated by one of skill in the art in possession of the present disclosure, prior to operating the fan modules **500** to cool the networking device **300**, the fan control operations **900** may include fan module detection and programming operations that detect and program each of the fan modules **500** in a similar manner (regardless of whether they are directly coupled to the motherboard **308** or coupled to the motherboard **308** via the COFTA system **200**).

[0043] As described above, in embodiments in which the circuit board **211** for the compute subsystem **210** is mounted to the fan chassis **202** via the plurality of stand-offs **211***a* to provide the circuit board **211** in the spaced-apart orientation along the Z-axis from the adjacent surface of the fan chassis **202** as illustrated in FIG. **2**B, additional thermal cooling may be provided for the compute subsystem **210** by allowing the airflow produced by the fan modules **500** to move past opposing surfaces (e.g., the top and bottom surfaces in FIG. **2**B) of the circuit board **211**. Furthermore, as discussed briefly above and as will be appreciated by one of skill in the art in possession of the present disclosure, the fan controller connector **208** on the COFTA system **200** may be configured to aggregate signals from the fan modules **500** coupled to the COFTA system **200** and provide the aggregated signal to the fan controller subsystem **316**, while allowing a signal received from the fan controller subsystem **316** to be provided to each of the fan modules **500** coupled to the COFTA system **200** as well. However, while the aggregation of signals from fan modules **500** on the COFTA system **200** and the provisioning of an aggregated signal to the fan controller subsystem **316** on the networking device **300** (and the splitting of a signal from the fan controller subsystem **316** between those fan modules **500**) has been described, one of skill in the art in possession of the present disclosure will appreciate how the transmission of separate signals between the fan modules **500** on the COFTA system **200** and the fan controller subsystem **316** will fall within the scope of the present disclosure as well.

[0044] The method **400** then proceeds to block **408** where a compute subsystem in the COFTA system performs compute functionality for the networking device. With reference to FIG. **10**, in an embodiment of block **408**, the compute subsystem **210** may perform compute functionality operations **1000** to provide any of a variety of compute functionality for the networking device **300**. For example, the compute functionality operations **1000** may include CPU **212***a* providing a Network Operating System (NOS) for the networking device **300** by booting up via a BIOS provided using the SPI flash storage device **212***b*, and providing the NOS following boot using the SSD storage device **212***c* and the DRAM device(s) **212***d*. However, while specific compute functionality including providing an NOS is described, one of skill in the art in possession of the present disclosure will appreciate how any of a variety of compute functionality (e.g., security compute functionality using the TPM device **212***f*) will fall within the scope of the present disclosure as well.

[0045] The method **400** then proceeds to decision block **410** where the method **400** proceeds depending on whether access to the compute subsystem is required. Similarly as discussed above for conventional COM Express devices, the compute subsystem **210** on the COFTA system **200** may require servicing or replacement for a variety of reasons that include errors associated with any of the processing systems (e.g., CPU **212***a* and/or the CPLD **212***g*), degradation of any of the storage device(s) (e.g., the SPI flash storage device **212***b* and/or the SSD storage device **212***c*), issues with the memory device(s) (e.g., the DRAM device(s) and/or the EEPROM device **212***e*), issues with the security device (e.g., the TPM device **212***f*) and/or other compute subsystem issues that would be apparent to one of skill in the art in possession of the present disclosure.

[0046] Furthermore, in some situations, the upgrade (or downgrade) of the compute subsystem **210** on the COFTA system **200** may be desirable, such as when the NOS requirements for the networking device **300** have changed, storage requirements for the networking device **300** have changed, processing requirements for the networking device **300** have changed, security requirements for the networking device **300** have changed (e.g., an upgraded TPM device is required to address evolving Federal Information Processing Standard (FIPS) requirements), etc. As such, at decision block **410**, the method **400** may proceed depending on whether the compute subsystem **210** requires service, replacement, upgrade, or downgrade, as well as any other reason for compute subsystem access that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, one of skill in the art in possession of the present disclosure will appreciate how access to the compute subsystem **210** may be required if the fan controller connector **208** on the COFTA system **200** requires service, replacement, upgrade, or downgrade as well.

[0047] If, at decision block **410**, access to the compute subsystem is not required, the method **400** returns to block **406**. As such, the method **400** may loop such that the fan modules **500** are operated to cool the networking device **300** (when needed) similarly as described above with reference to block **406**, and the compute subsystem **210** in the COFTA system **200** performs compute functionality for the networking device **300** (when needed) similarly as described above with reference to block **410**, as long as access to the compute subsystem **210** is not required.

[0048] If at decision block **410**, access to the compute subsystem is required, the method **400** proceeds to block **412** where the COFTA system is decoupled from the networking device. In an embodiment, at block **412** an in situations in which access to the compute subsystem **210** is required, the COFTA system **200** may be removed from the networking device **300**. In some embodiments, prior to removing the COFTA system **200** from the networking device **300** (and thus prior to decoupling the compute subsystem **210** from the motherboard **308** in the networking device and, in turn, from the network processing system **318**), a network administrator or other user of the networking device **300** may use the NOS provided by the compute subsystem **210** to instruct the network processing system **318** (e.g., a data plane provided by the network processing system **318**) to operate in a "compute detached mode" that configures the network processing system **318** (e.g., the data plane provided by the network processing system **318**) to continue to perform networking functionality (e.g., data routing and/or other active data plane operations known in the art) when the compute subsystem **210** is decoupled from the networking device **300**.

[0049] In addition, in some embodiments, the "compute detached mode" may cause the network processing system **318** to instruct the fan controller subsystem **316** to increase the speed one or more of the fan module(s) **500** coupled to the fan module connectors **310***a*-**310***d* to compensate for the loss of the fan modules **500** coupled to the COFTA system **200** when the COFTA system **200** is removed from the networking device **200**. However, while two specific operations have been described that may be performed when the COFTA system **200** is disconnected from the networking device **300**, one of skill in the art in possession of the present disclosure will appreciate how other operations may be performed to compensate for the disconnection of the COFTA system **200** from the networking device **300** while remaining within the scope of the present disclosure as well.

[0050] As will be appreciated by one of skill in the art in possession of the present disclosure, the COFTA system **200** may be removed from the networking device **300** may grasping the COFTA system **200** (or the handles included on the fan modules **500** that are coupled to the COFTA system **200**, visible in FIG. 5B) and pulling the COFTA system **300** such that the fan controller connector

208 and the networking device connector **214** on the compute subsystem **210** disengage the fan subsystem connector **312** and the compute subsystem connector **314** on the motherboard **308**, respectively, to decouple the compute subsystem **210** from the network processing system **318**, allowing the COFTA system **300** to move out of the COFTA channel **308***a* defined by the motherboard **308**, and out of the COFTA slot **306** defined by the device chassis **302**.

[0051] As will be appreciated by one of skill in the art in possession of the present disclosure, the removal of the COFTA system **200** from the networking device **300** allows for servicing, replacement, upgrade, or downgrade of the compute subsystem **210** (or the fan controller connector **208**) on the COFTA system **200**. For example, in situations in which the compute subsystem **210** (or the fan controller connector **208**) on the COFTA system **200** requires service, such service may be performed "in the field" (i.e., at the location of the networking device **300**) after removing the COFTA system **200** from the networking device **300**, or may simply require the shipping of the COFTA system **200** to an authorized service entity (e.g., with a temporary replacement COFTA system provided for the networking device **300** while the COFTA system **200** is being serviced in some situations), rather than requiring the shipping of the entire networking device **300** as is required in conventional systems. Similarly, the replacement, upgrade, or downgrade of the compute subsystem **210** (or the fan controller connector **208**) on the COFTA system **200** system may simply require the replacement COFTA system, upgraded COFTA system, or downgraded COFTA system to be shipped to the location of the networking device **300**, and swapped out with the COFTA system **200** similarly as described above.

[0052] Furthermore, as discussed above, the networking device **300** may remain in its rack and may be configured to continue to perform networking functionality while the COFTA system **200**/compute subsystem **210** is decoupled from the networking device **300**, and may also operate to increase the speed of any of the fan module(s) **500** coupled to the fan module connectors **310***a*-**310***d* (e.g., by doubling the speed of those fan modules **500**) to compensate for the removal of the fan modules **500** that are coupled to the COFTA system **200** and ensure that any components in the networking device **300** are adequately cooled while the COFTA system **200** is removed from the networking device **300**.

[0053] Thus, systems and methods have been described that provide a Compute-On-Fan-Tray-Assembly (COFTA) system that includes a compute subsystem mounted to a fan chassis that may be easily coupled to, and decoupled from, a networking device. For example, the COFTA system of the present disclosure may include a fan chassis that is configured to be positioned in a networking device, and that has a fan module coupling subsystem that is configured to couple at least one fan module to the fan chassis. A compute subsystem is mounted to the fan chassis and is configured to perform compute functionality for the networking device. A networking device connector is included on the fan chassis, is coupled to the compute subsystem, and is configured to engage a compute subsystem connector in the networking device to couple the compute subsystem to a network processing system in the networking device when the fan chassis is positioned in the networking device. As described above, embodiments of the COFTA system of the present disclosure allow the compute subsystem for the networking device to be connected to, and disconnected from, the networking device without the need to remove the networking device from its rack or suspend networking functionality performed by the network processing system in the networking device, eliminating issues with servicing and/or replacement of such compute subsystems, and enabling the upgrading (or downgrading) of such compute subsystems.

[0054] Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

Claims

1. A Compute-On-Fan-Tray-Assembly (COFTA) system, comprising: a fan chassis that includes a fan module coupling subsystem that is configured to couple at least one fan module to the fan chassis, wherein the fan chassis is configured to be positioned in a networking device; a compute subsystem that is mounted to the fan chassis and that is configured to perform compute functionality for the networking device; and a networking device connector that is included on the fan chassis, that is coupled to the compute subsystem, and that is configured to engage a compute subsystem connector in the networking device to couple the compute subsystem to a network processing system in the networking device when the fan chassis is positioned in the networking device.
2. The system of claim 1, wherein the fan module coupling subsystem is configured to couple a plurality of fan modules to the fan chassis.
3. The system of claim 1, wherein the compute subsystem includes a Central Processing Unit (CPU) that is configured to provide a Network Operating System (NOS) for the networking device.
4. The system of claim 1, wherein the networking device connector is configured to engage the compute subsystem connector in response to the fan chassis being moved into the networking device.
5. The system of claim 1, further comprising: a fan controller connector that is included on the fan chassis, that is coupled to at least one fan module connector that is included on the fan chassis, and that is configured to engage a fan subsystem connector in the networking device to couple the at least one fan module connector to a fan controller subsystem in the networking device when the fan chassis is positioned in the networking device.
6. The system of claim 1, further comprising: a plurality of standoff elements that mount the compute subsystem to the fan chassis to position the compute subsystem in a spaced-apart orientation from the fan chassis.
7. An Information Handling System (IHS), comprising: a networking device chassis that defines at least one fan slot; a network processing system that is housed in the networking device chassis and that is configured to perform networking functionality; a fan chassis that includes at least one fan module and that is positioned in the at least one fan slot defined by the networking device chassis; a compute subsystem that is mounted to the fan chassis and that is configured to perform compute functionality; and a networking device connector that is included on the fan chassis, that is coupled to the compute subsystem, that engages a compute subsystem connector in the networking device chassis to couple the compute subsystem to the network processing system, and that is configured to disengage the compute subsystem connector when the fan chassis is removed from the networking device chassis to decouple the compute subsystem from the network processing system.
8. The IHS of claim 7, wherein the fan chassis includes a plurality of fan modules.
9. The IHS of claim 7, wherein the compute subsystem includes a Central Processing Unit (CPU) that is configured to provide a Network Operating System (NOS) for the networking device.
10. The IHS of claim 7, wherein the networking device connector is configured to engage the compute subsystem connector in response to the fan chassis being moved into the networking device.
11. The IHS of claim 7, further comprising: a fan controller connector that is included on the fan chassis, that is coupled to the at least one fan module, that engages a fan subsystem connector in the networking device chassis to couple the at least one fan module to a fan controller subsystem in the networking device chassis, and that is configured to disengage the fan subsystem connector when the fan chassis is removed from the networking device chassis to decouple the at least one fan module from the fan controller subsystem.

12. The IHS of claim 7, further comprising: a plurality of standoff elements that mount the compute subsystem to the fan chassis to position the compute subsystem in a spaced-apart orientation from the fan chassis.
13. The IHS of claim 7, wherein the fan chassis includes at least one first fan module and is positioned in at least one first fan slot defined by the networking device chassis, and wherein at least one second module is positioned in at least one second fan slot defined by the networking device chassis.
14. A method for providing a compute system in a networking device, comprising: moving, by a fan chassis that includes a compute subsystem and at least one fan module, into a networking device; engaging, by a networking device connector on the fan chassis that is coupled to the compute subsystem and in response to the fan chassis moving into the networking device, a compute subsystem connector in the networking device to couple the compute subsystem to a network processing system in the networking device; performing, by the compute subsystem while the compute system is coupled to the network processing system, compute functionality for the networking device; moving, by the fan chassis, out of the networking device; and disengaging, by the networking device connector in response to the fan chassis moving out of the networking device, the compute subsystem connector to decouple the compute subsystem from the network processing system.
15. The method of claim 14, wherein the fan chassis includes a plurality of fan modules.
16. The method of claim 14, wherein the compute subsystem performing compute functionality for the networking device while the compute system is coupled to the network processing system includes: providing, by a Central Processing Unit (CPU) in the compute subsystem while the compute system is coupled to the network processing system, a Network Operating System (NOS) for the networking device.
17. The method of claim 14, further comprising: engaging, by a fan controller connector on the fan chassis that is coupled to the at least one fan module and in response to the fan chassis moving into the networking device, a fan subsystem connector in the networking device to couple the at least one fan module to a fan controller subsystem in the networking device.
18. The method of claim 17, further comprising: disengaging, by the fan controller connector in response to the fan chassis moving out of the networking device, the fan subsystem connector to decouple the at least one fan module from the fan controller subsystem.
19. The method of claim 14, wherein the compute subsystem is mounted to the fan chassis via a plurality of standoff elements that position the compute subsystem in a spaced-apart orientation from the fan chassis.
20. The method of claim 14, wherein the fan chassis includes at least one first fan module and is positioned in at least one first fan slot defined by the networking device chassis, and wherein at least one second fan module is positioned in at least one second fan slot defined by the networking device.